

Smart Rail, New Pulse: The Jing-Zhang Ren-Line Century AI Innovation Belt Overall Urban Design

Using the Jing-Zhang Railway herringbone alignment as a spatial motif, the proposal presents one north-south spine, two wings, three key areas, three pilgrimage landmarks, and 12 AI scenario cards. V3 adds an independent jury first-read booklet, a four-part heritage–service–governance rule, a conceptual active-travel anchor, an AI scenario-station service blueprint, inclusive user journeys, and exit-ready decision gates. All geometry derives from the provisional boundary and is recalculated in EPSG:4548; digital service uses no real personal data, and the full package is recomputed when formal data are published.

阅读中文版本

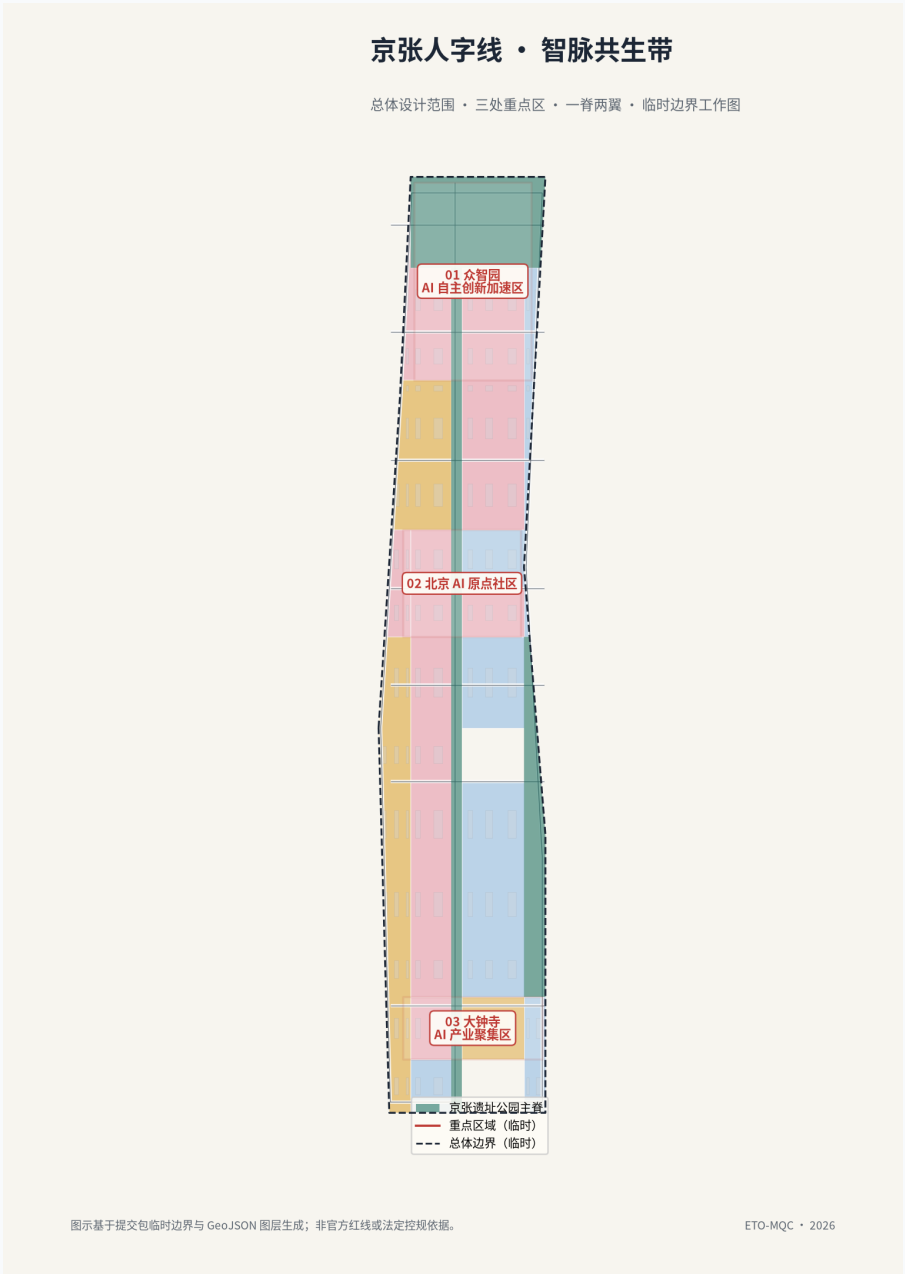
Smart Rail, New Pulse: The Jing-Zhang Ren-Line Century AI Innovation Belt Overall Urban Design

Design Basis and Source List

This formal proposal takes the Beijing Municipal Planning and Natural Resources Commission Haidian Branch prequalification announcement for the Century Jing-Zhang AI Innovation Belt International Urban Design Solicitation as its primary basis, and uses the provisional boundary, key areas, enums, metrics, and source registry maintained in `brief/site-package/` as machine-readable evidence . The AI agent read `design_brief.json` , `allowed_design_space.json` , `sources.json` , `enums/` , `ranges/` , `schemas/` , `data/source_registry.json` , and `data/processed/agent_fact_pack.md` , then used `project_scope_summary.csv` , `agent_task_requirements.csv` , `source_use_matrix.csv` , and `missing_data_checklist.csv` to build task, scope, usage, and gap lists .

Usage boundaries follow `data/source_registry.json` : the agent must not upgrade background-only or provisional-only materials into official boundary, statutory control plan, scoring basis, or government implementation commitments. The provisional boundary and three key-area polygons come from `brief/site-package/geometry/provisional_boundaries.geojson` and are used for generation, self-check, and visualization only . This proposal is not a standalone vision document; every spatial conclusion returns to a GeoJSON layer, every performance conclusion returns to a metrics table, and professional judgment returns to the standard matrix and design depth matrix .

The evaluable status of this generation is: provisional boundary, with precision warnings retained and recalculation required after official data release; this does not block content scoring. Spatial boundaries trace back to the site boundary layer and area metrics , and the three key areas are cross-checked by their own layers and count metric .

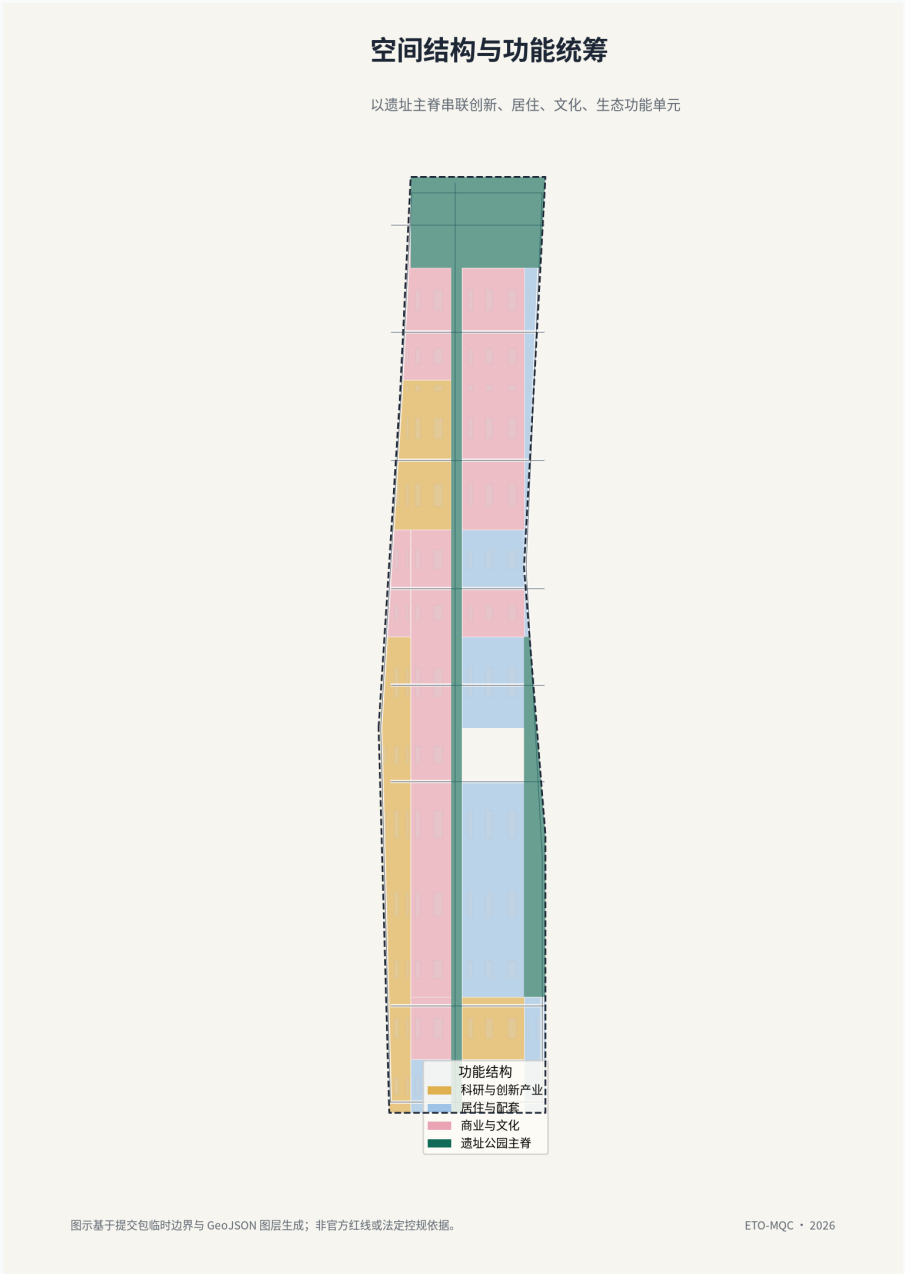


Evidence chain overview

Three-Level Scope Framework

The scheme organizes work in three levels defined by the announcement : the coordinated research area covers about 43.6 km2 for AI industry ecosystem, strategy, innovation chains, and future city form; the overall design area covers about 11.4 km2 around the Jing-Zhang Heritage Park for urban renewal framework, industrial space layout, transport and municipal support, and urban character control ; the key areas cover about 369 ha for three detailed design districts, requiring functional uses, building scale, retain-renovate-demolish classification, public space connectivity, and transport organization . The three levels are mapped item by item in `compliance_matrix.json` so every mandatory task under announcement items 1.3, 1.4, 1.5 and agent.1-agent.6 has section, layer, metric, drawing, and HTML evidence .

Level	Design question	Proposal answer	Data anchor
Coordinated research	How to organize AI ecosystem and future city form	Innovation chain from universities to open source to enterprises to public experience to global outreach, with the Ren-Line spatial motif	compliance_matrix.json, standard_matrix.json
Overall design	How to map industry, renewal, transport, and character	Land use, buildings, roads, green space, public space, and phasing layers	,
Key areas	How to reach detailed design depth	Positioning, spatial actions, AI scenarios, and implementation dependencies per district	, ,



Land use and spatial structure

The three levels are not disconnected drawing sets. Coordinated research shapes industry chain and city form judgment, overall design lands that judgment into renewal projects, spatial structure, and facility capacity, and detailed design verifies implementability at plot, building, transport, public space, and AI scenario scale. The agent fixed the provisional boundary and constraints before generating land use, buildings, roads, green space, public space, phasing, and AI service nodes, then recalculated metrics and flagged which conclusions remain limited by the provisional boundary .

Coordinated Research Area: Industry and Future City Research

The coordinated research area, roughly 43.6 km², is the first design level, focused on industry and future city research . Industry research answers how Haidian AI industry should organize its innovation chain and how the world-class AI innovation ecosystem required by the agent open call should land; future city research asks how AI will change work, life, social interaction, learning, transport, and public services. The proposal organizes an innovation chain of university ideation, open-source collaboration, enterprise transformation, public experience, and global outreach, and overlaps it with the Ren-Line spatial structure: the spine carries ideation and showcase functions, the wings carry technology services (west) and scenario enablement (east), and the three key areas carry indigenous innovation, talent origin, and AI-native industries .

The world-class AI innovation ecosystem maps to differentiated positioning across the three key areas and innovation service facilities along the belt: Zhongzhi Park hosts full-stack indigenous innovation plus governance discourse, AI Origin Community hosts ecosystem and talent origin, and Dazhongsi hosts AI-native industries plus international outreach . The three districts total about 369 ha , served by 8 AI scenario stops and 12 AI scenario cards , validated against five user personas and anchored in public space, green space, and road layers . Statements about global AI activities, developer communities, open scenarios, or pilgrimage routes are marked as conceptual proposals or reference options for professional teams, not as confirmed government activities or implementation arrangements .

The concept identity is **Jingzhang Herringbone Line • Living Intelligence Spine**. Reviewable logo, bilingual wordmark, colour, wayfinding, pass, digital-platform, and annual-event prototypes are supplied under [assets/identity/](#). The mark uses the Jing-Zhang railway herringbone as its geometry: Rail Blue carries the engineering heritage, Spine Green the continuous civic ecology, Signal Orange the visible AI-service interface, and Node Red the AI public node that requires human confirmation. It is not a government emblem, registered trademark, or authorized partner brand; any public implementation requires trademark search, wayfinding approval, and rights clearance .



京张人字线

智脉共生带

JINGZHANG HERRINGBONE LINE

LIVING INTELLIGENCE SPINE

标志语义：铁路人字展线 • 智能节点 • 生态公共主脊

Meaning: rail herringbone · intelligence node · civic ecological spine

概念性品牌识别资产 | 用于方案呈现；不主张已注册商标或官方标识。

Jingzhang Herringbone Line logo, bilingual wordmark, and meaning

Identity element	Deliverable and rule	Use boundary
Mark and wordmark	logo-lockup.svg/png ; the bilingual lockup reads “京张人字线 • 智脉共生带 / Jingzhang Herringbone Line • Living Intelligence Spine.”	Concept asset only; it does not confer official identity or trademark rights.
Palette	Rail Blue #274E88 , Spine Green #126A58 , Signal Orange #D9822B , Node Red #BE3A34 , Warm Paper #F7F5EF .	Keep high contrast in wayfinding, furniture, pages, and event material; never encode safety information by colour alone.
Wayfinding and access	Bilingual AI Scenario Station prototype includes human assistance, accessibility, and real-time information.	Requires transport, accessibility, and advertising approval before deployment.
Digital and event applications	Spine Pass, scenario-opening platform, and annual AI week visual prototypes.	No individual-route collection; event names, partner marks, and data interfaces require separate authorization.

京张人字线 · 智脉共生带

视觉识别概念系统 | Logo / 色彩 / 导视 / 场景应用

JINGZHANG HERRINGBONE LINE
Living Intelligence Spine

01 · 标志构图



“人”形轨迹转译创新连接；中心节点代表公共 AI 场景接口。

02 · 标准色



Rail Blue
#274E88



Spine Green
#126A58



Signal Orange
#D9822B



Node Red
#BE3A34



Warm Paper
#F7F5EF

03 · 导视原型

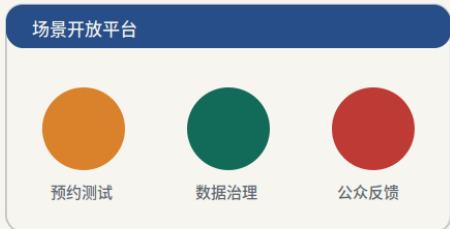


双语、色弱友好、保留非数字化服务入口。

04 · 应用原型



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Concept VI: mark, colour, wayfinding, and scenario applications

The six global cases below are mechanism references, not confirmed partnerships, endorsements, or rights grants. The Toronto, Seoul, and Singapore cases are registered in [sources.json](#) and [report/narrative.md](#) with publisher, linked source, date information, transferable mechanism, and use boundary .

London, Qianhai, and Shibuya are likewise verifiable mechanism comparisons. The submission does not reproduce their photographs, logos, drawings, dashboards, or copyright-protected text .

Case	Publisher, source, and date	Transferable mechanism	Licence and applicability boundary
Toronto Quayside	Waterfront Toronto project page; verified 2026-08-12.	Coordination among public-realm delivery, competitive development partner, and walkable waterfront.	Linked and originally summarized only; no renderings, logos, or data used.
Seoul Smart City & Digitization	Seoul Metropolitan Government, Smart City & Digitization Master Plan (2021–2025); verified 2026-08-12.	Public-benefit, inclusion, cyber-safety, and participation principles.	Not a replication of DMC; Seoul retains copyright and no visual material is used.
Punggol Digital District	JTC Corporation; webpage updated 2026-01-14.	Industry-academia proximity, community integration, district operating-data platform, and low-carbon mobility.	Mechanism comparison only; no JTC plan or photography redistributed.
London Knowledge Quarter	University College London; 2024–2025 dashboard materials.	Multi-institution network plus explainable research and industry performance dashboards.	UCL/Elsevier require attribution for findings; no data or screenshots redistributed.
Qianhai Shenzhen-Hong Kong Cooperation Zone	Shenzhen Municipal Commerce Bureau; 2024-08-08.	Cross-regional institutional, transport, and ecological interface coordination.	Page prohibits republication without written authorization; no claim of Qianhai partnership.
Tokyo Shibuya Stream	Tokyo Convention & Visitors Bureau; 2025-12-11.	Former-railway redevelopment linked with river promenade, squares, and mixed-use facilities.	Page images carry copyright; no images, marks, or drawings reproduced.

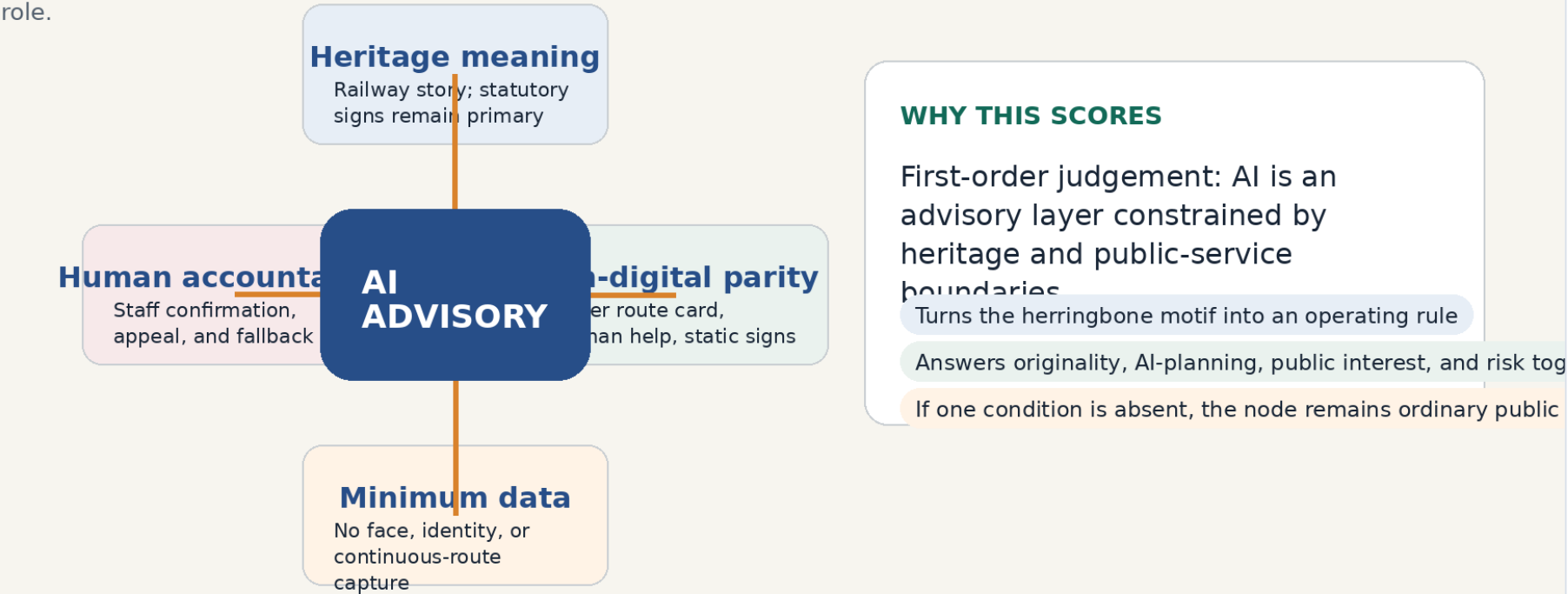
Applicability is this proposal's original interpretation of mechanisms. It requires revalidation against Haidian's official boundaries, regulations, property rights, public engagement, and professional advice .

V3 Jury First-Read Rule: Four-Part Heritage–Service–Governance Transformation

V3 translates the Jing-Zhang Ren-Line from a visual motif into a reviewable public-service rule. Every place described as an AI public-service node must jointly provide **heritage meaning, non-digital parity, minimum data, and human accountability**. Heritage meaning takes priority over decorative technology; paper guides, static signs, and human assistance keep basic information available to visitors who do not use digital tools; any digital function handles only non-personal input necessary for its purpose; and site staff, professional responsibility, and statutory signs prevail over algorithmic advice. If any one condition is absent, the place remains ordinary public space and must not be packaged as a deployed smart scenario.

Heritage is not a backdrop: it becomes a rule for accessible, interpretable, exit-ready AI public service

Every node combines heritage interpretation, non-digital parity, data minimisation, and a human accountable role.



Rule diagram · All spatial and service claims remain conceptual; recompute after formal data release.

ETO-MQC · ai-heritage-spine · concept evidence

Four-part heritage-service-governance transformation rule

The V3 primary scenario is a low-risk conceptual prototype for heritage-walk guidance and an AI scenario station—not autonomous driving, public surveillance, or a real-data system. Visitors may use a paper card, bilingual signs, human assistance, or an optional digital interface to select guidance preferences such as fewer steps, shade, heritage interpretation, or family rest. Digital advice may only read human-verified public-facility status; it must not collect face data, identity, or continuous personal routes. Staff and site signs make final calls on temporary closure, safety messages, and route opening. Stale data, anomalous complaints, or safety incidents require digital advice to switch off and fall back to static signs, human assistance, and site checks. This scenario defines service and governance boundaries only; it makes no claim of accuracy, response time, real collection capability, or approved deployment .

Heritage-walk guidance and AI scenario station: algorithm advises; people carry public responsibility

Every step has a minimum-data boundary, non-digital parity, and an explicit exit route.

Arrive	Choose	Advise	Confirm	Feedback
Bilingual overview paper route card human-help bell No sign-in no identity	Shade / fewer steps heritage / family rest Temporary preference no profile	Explainable route barrier / rest prompt Public facility status no route capture	Staff confirm closures site signs prevail Algorithm makes no safety call	Paper / in-person / phone accessible feedback route Themes only no marketing

FAIL-SAFE / HUMAN FALLBACK

Abnormality / stale data / complaint / safety event → switch off digital advice → static signs + human help + site check

Service blueprint · No real personal data; automation never replaces public decisions, site signs, or human service.

ETO-MQC · ai-heritage-spine · concept evidence

AI scenario station service blueprint and human fallback

Overall Design Area: Urban Renewal and Regulatory-Plan-Level Urban Design

The overall design area must reach regulatory-plan-level urban design depth . The proposal presents an urban renewal spatial framework, low-efficiency space identification, a renewal project list, implementation policy recommendations, industrial function ratios, spatial organization patterns, total building scale, and carrying capacity assessment. `geometry/land_use.geojson` fully and gaplessly covers the design boundary ; `geometry/buildings.geojson` expresses 104 conceptual building footprints ; `geometry/roads.geojson` expresses micro-circulation, slow-traffic, and rail transfer relations ; and `metrics.json` recalculates core areas and ratios .

Regulatory-plan content is decomposed into reviewable objects: land use structure, building footprints, transport organization, green and public space ratios, phasing scope, and AI scenario nodes. Content touching building height, development intensity, road red lines, setbacks, and facility standards is written as pending confirmation by official regulatory conditions where no official control data exists, and agent-guessed values are never presented as approved indicators ; related depth items are and . The overall design must support transport, rail, municipal, and supporting facilities: rail-station integrated development, road micro-circulation, non-motorized parking, parking supply, innovation service platforms, talent and life services, new infrastructure, distributed energy, and edge computing are proposed as spatial layouts and implementation paths , while `geometry/phasing.geojson` expresses the three-phase implementation scope .

AI Innovation Ecosystem, Personas, and AI+ Scenarios

The AI innovation ecosystem organizes space and facilities around full-stack indigenous innovation, a world-class ecosystem, and scenario enablement. Five user personas (open-source developers, startups, enterprise visitors, surrounding residents, and university students) are translated into locatable spatial responses. AI+ scenarios are placed along the spine and wings in

two categories: industry-development scenarios and AI-empowered urban-function scenarios . Personas and scenario cards are listed below, each anchored in public space, green space, and road layers ; counts are checked by the scenario card and persona indicators .

The three AI industry test scenarios below are **conceptual pilot cards**, not connected data systems, approved road tests, or government commitments. Their thresholds are suggested pilot evaluation targets. Before activation, responsible authorities, operators, and data subjects must confirm lawful basis, information security, accessibility, and human-review procedures .

Test scenario	Data inputs and minimization boundary	TRL / test metric	Spatial component	Suggested operator	Exit condition and human fallback
Automated active-travel navigation	Authorized high-definition map, anonymized flow and traffic conditions; no identifiable face or continuous individual-route data.	TRL 6; navigation-advice accuracy target $\geq 95\%$, end-to-end prompt latency target ≤ 200 ms.	Jing-Zhang Heritage Park active-travel spine; conceptual service extent about 8 km.	Scenario operator, transport coordinating party, and accessibility representatives.	Pause automated prompts after failed target, abnormal complaints, or safety incident; revert to static wayfinding, human assistance, and on-site patrols.
AI public-space maintenance	Facility-state sensors, de-identified/cropped inspection images, and human maintenance tickets; no resident profiling.	TRL 5; fault-detection target $\geq 90\%$, maintenance-response target ≤ 30 min.	Qinghe waterfront park, station plazas, and AI scenario stations.	Parks team, municipal maintenance team, and site operator.	Shrink deployment when cost exceeds budget, false positives become unacceptable, or maintenance burden is high; manual inspection remains the primary process.
Data-compliance circulation sandbox	De-identified training samples, authorization register, and tamper-evident audit log; no unauthorized production data.	TRL 4; zero confirmed leakage events, audit-completeness target 100%.	Zhongzhiyuan data-compliance circulation node (concept indoor/service space).	Data controller, security-governance team, and independent compliance reviewer.	Freeze processing, preserve audit records, and conduct human review if law, consent purpose, or security baseline changes; no automation substitutes legal judgment.

Persona	Typical needs	Spatial response	Review boundary
Open-source developers	Release, collaboration, testing, reputation	Origin community release hall, public code wall, overnight collaboration space	No personal behavior tracking; aggregate statistics only
Startups	Low-cost offices, compute access, product testbed	Zhongzhi Park shared test field, edge compute service points, governance consulting	Compute and data services require separate authorization
Enterprise visitors	Showcase, business, international reception, recruitment	Dazhongsi international roadshow lounge, transit connections, public space near anchors	Company marks and cases require rights clearance
Residents	Commute, leisure, community services, low-disruption renewal	Heritage Park slow-traffic loop, embedded community services, graded night lighting	No resident profiling for commercial recommendation
University students	Commercialization, cross-campus work, daily slow travel	Campus-park slow connections, commercialization stops, AI education experience points	Campus data and research results require authorization

Scenario card	Spatial carrier	Design note
01 Open-source release hall	AI Origin Community	Release, code contribution showcase, and small roadshows for universities, open-source communities, and startups
02 Safety governance sandbox	Zhongzhi Park	Translates standard-setting, safety evaluation, and red-team testing into visitable, bookable, regulated nodes
03 Edge compute stop	Overall design area	New-infrastructure prototype tied to public services and distributed energy, pending deepening
04 AI slow-traffic navigation	Heritage Park vitality belt	Explainable wayfinding and low-intrusion sensing to identify gaps, congestion, and accessibility needs
05 Dazhongsi international roadshow lounge	Dazhongsi	Showcase, negotiation, press, and international exchange for agents, devices, and content enterprises
06 Qinghe low-carbon innovation corridor	Zhongzhi Park riverside	Combines green space, stormwater, walking and cycling with AI display as district public living room
07 Near-campus commercialization street	AI Origin Community	Incubation, showcase, legal, IP, and investment services for university commercialization
08 Data-elements parlor	Dazhongsi	City-service interface for data-element and digital-asset circulation under compliance and audit
09 AI life-services sample street	Community-commerce junctions	AI+ healthcare, education, legal, and life services in operable small-block spaces
10 Global AI event week route	Belt public space system	A walkable, shareable experience from heritage culture to open source to industry to roadshow
11 Origin pilgrimage plaza	AI Origin Community	Memorial, release, and public-art landmark at the Ren-Line vertex
12 Dazhongsi station-city living room	Dazhongsi front	Integrated transit, retail, exhibition, and digital-experience complex

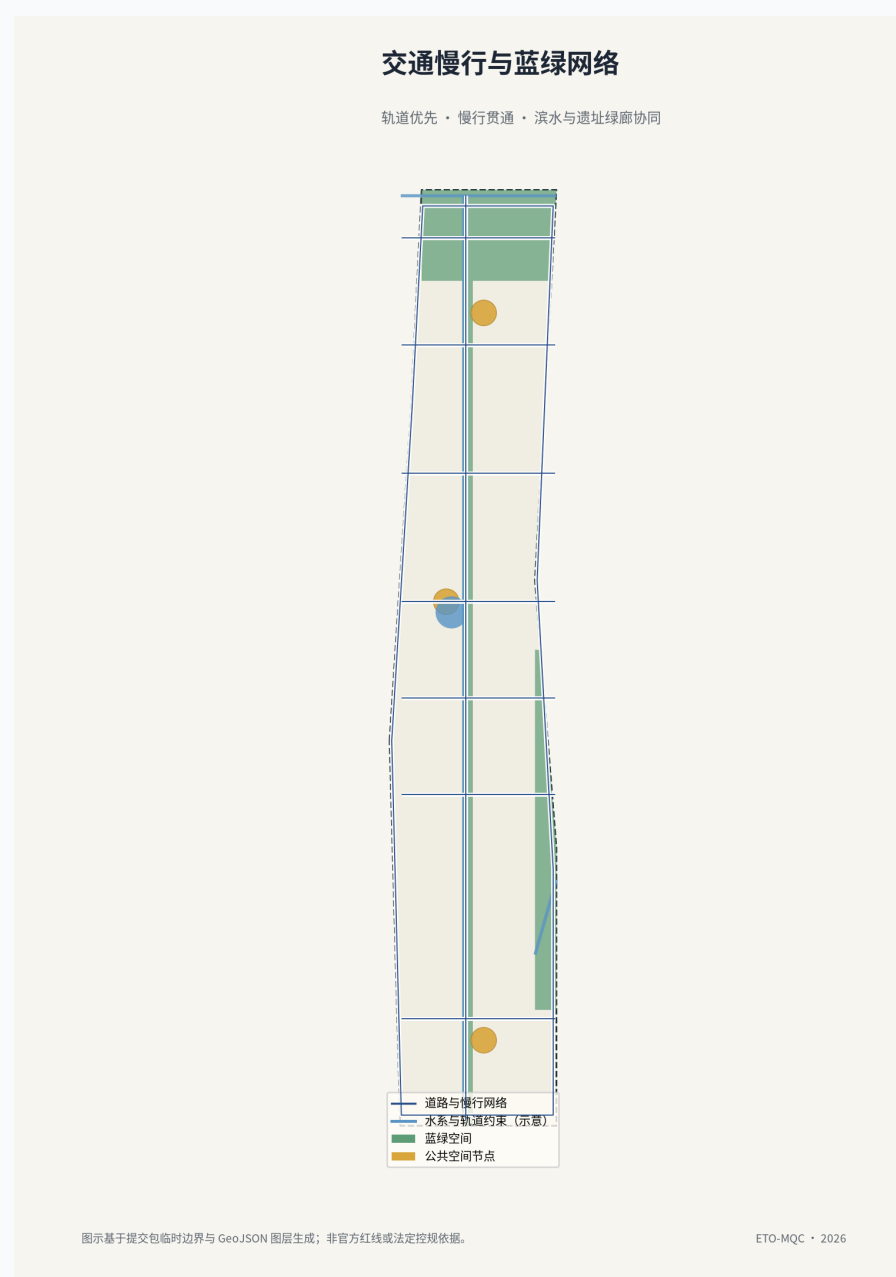
Land Use, Building Scale, and Retain-Renovate-Demolish Strategy

Land use uses `geometry/land_use.geojson` as the machine-readable base map, cut on a grid into gapless parcels within the design boundary and classified by `enums/land_use_codes.json`, organizing research, commercial service, education, culture, green, and plaza functions; layout depth is covered by . Building scale is expressed as 104 conceptual building footprints totaling about 869,111 m², following `enums/building_types.json`. Because floor area ratio, total floor area, building height, and density lack official regulatory data, the proposal does not compile formal building scale indicators from guessed values; related depth items are and .

The retain-renovate-demolish strategy follows a retain-first, low-disruption renewal principle : the Heritage Park, Qinghe riverside green space, railway heritage, and historical remains are retained first ; the building footprint layer expresses an indicative retain/renovate/demolish classification, with actual action subject to property rights and implementation conditions . The renewal project list and phasing are covered in their own sections, and all land and building-scale conclusions return to `metrics.json` so recalculation stays traceable .

Transport, Rail, Municipal Infrastructure, and Public Services

Transport follows a rail-first, slow-traffic-first, hierarchical network principle: rail stations form TOD nodes with seamless transfer among rail, bus, walking, and cycling; roads are layered into arterials, collectors, local roads, and slow-traffic-priority routes, with emphasis on closing the east-west slow-traffic gaps across the Heritage Park spine ; depth is covered by . The three key areas organize station-city integration around Dazhongsi station, the Origin Community station, and the park hub respectively . The slow-traffic system runs along the spine and wings, linking 8 AI scenario stops and three pilgrimage landmarks .



Rail, slow traffic, and blue-green systems

Municipal and public services are arranged in two levels under TOD priority: district facilities concentrate around stations, and community facilities embed into residential and industrial clusters . New infrastructure (edge compute, smart poles, smart sanitation, energy, and reclaimed water) is proposed as prototypes to be deepened with municipal, power, and telecom special plans. Rail alignments, stations, and high-voltage corridors follow the constraint layer; the proposal does not change existing rail red lines or municipal corridors . All transport and municipal conclusions derive from the provisional boundary and public data and await official data confirmation .

Blue-Green Network, Public Space, and Urban Character

The blue-green network takes the Heritage Park as the north-south spine, linking Qinghe and Xiaoyue River waterfronts and parkland into a one-spine-two-wings, interwoven open-space skeleton ; the green ratio is checked by . Public space follows a node-axis-venue structure: station plazas, AI scenario stops, and pilgrimage landmarks form activity nodes; the spine slow-traffic system forms the axis; parks and plazas form open venues ; the public-space ratio is checked by , with depth in . Public space emphasizes accessibility, continuity, barrier-freedom, and all-age friendliness, connected by slow-traffic paths.

Public-space component library	Core provision	Data and operation boundary
AI Scenario Station	Bilingual wayfinding, bookable test bench, human assistance, accessible seating, paper guide, and low-power information screen.	Real-time information is optional; paper, static signs, and human assistance preserve basic service offline.
Heritage active-travel corridor	Railway-remnant viewing band, continuous shade, cycle stop, low-level night guidance, and rain garden.	It does not alter heritage or rail-safety boundaries; lighting and sensors require heritage, transport, and electrical review.
Station-front shared living room	Canopy, waiting seats, accessible ramp, movable exhibition stand, and child/carer facilities.	Fire safety, passenger flow, accessibility, and advertising permits are prerequisites.
Community micro-renewal pocket	Rest, exercise, family, community notice, and non-digital service point.	Community co-creation determines content; no resident profile is used for marketing or automated decisions.

Landmark catalogue	Location and form	Recognition and display system
AI Origin Plaza	Herringbone-vertex public plaza in the Beijing AI Origin Community; release, public art, and remembrance.	Renewable displays for open-source contribution, public service, and accessible co-creation; recognition requires transparent rules and external review.
Jingzhang Intelligence Spine Bridge/Walk	Lightweight crossing and lookout on the heritage spine (concept).	Explains railway history, repair process, and open-data method; does not replace statutory heritage interpretation.
Dazhongsi Station-City Living Room	Transfer, exhibition, and digital-experience node at Dazhongsi station front.	Names, logos, work, and personal data require authorization before showing enterprise or university results.

The heritage-resource system follows six steps: identify, grade, protect, adaptively activate, interpret, and monitor. Official heritage listings, railway remnants, mature trees, water, and oral-history material first require inventory and property-right verification; protection grade and sensitivity then determine no-touch, low-impact renewal, and interpretable-use interfaces. Rail Blue, Spine Green, Signal Orange, and Node Red form bilingual wayfinding without covering statutory heritage signs. “See the rails, remember the history, feel the future” is a conceptual international-communication line whose content must be archival-verified, translation-reviewed, and rights-cleared before publication. This framework does not replace legal heritage recognition or protection requirements .

Urban character is themed as century-old Jing-Zhang coexisting with the AI future: heritage park and railway imagery are retained, innovation buildings are encouraged to adopt the Ren-Line roof and railway-motif language, and height and setback controls near rail, water, and heritage interfaces are managed to create a place where rails remain visible, history is remembered, and the future can be felt . Character controls (color, material, roof form, signage, nightscape) are proposed as design guidance to be deepened under formal urban design and regulatory conditions; at heritage-protection locations the proposal only expresses retain-and-revitalize intention without altering protection requirements .

V3 Concept Heritage-Walk Anchor and User Journeys

To make the overall strategy independently reviewable, V3 selects an **approximately 800 m conceptual working strip** from the spine rather than adding a statutory alignment, property line, or construction set-out. North heritage interpretation/ human help, middle shade/accessible rest, and south scenario station/fallback signs demonstrate a minimum space–service–governance loop. The working strip can only move to further professional analysis after space/heritage, data, operations, and public conditions have been reviewed; if any condition is absent, only paper guidance and human-service recommendations remain .

V3 ANCHOR · 02

CONCEPT ONLY · V3 JURY EDITION

Approx. 800 m concept heritage-walk anchor: from spatial continuity to service continuity

This is not a legal alignment, property line, or construction set-out; it is a reviewable working strip.

CONCEPT WORKING STRIP · NOT A REDLINE

01 North:
heritage meaning / human help
Paper route card · large-print guide · facility-status notice

02 Middle: shade / accessible rest
Continuous stop · rain garden · low-light orientation

03 South:
scenario station / fallback signs
Explainable advice · staff confirmation · works offline

Space / heritage gate
Formal extent, heritage impact, rail safety, fire and access conditions

Data gate
Public facility status, purpose limitation, retention; no real personal data

Operations gate
Staffing, upkeep, complaints and access review; no named operator, no pilot

Anchor diagram · Working extent only; without gate evidence, no delivery or public pilot proceeds.

ETO-MQC · ai-heritage-spine · concept evidence

Approximate 800 m conceptual heritage-walk anchor

Inclusion does not stop at abstract user labels; it is tested through a service journey that can be completed. An older visitor should obtain a large-print route, seating, and human guidance without a phone. A blind or mobility-limited visitor should be able to identify a continuous accessible route, alternatives, and risk notices. A digitally excluded visitor who declines all data entry should still receive basic information that is no worse than the information available to a digital user. These are design assumptions awaiting co-design and professional accessibility review; they do not replace field tests or statutory facility acceptance .

Three user journeys: declining digital service must still deliver equivalent basic information

Inclusion becomes a route to complete, a service point to find, and a question to answer—not a user label.

Older visitor Starting barrier Long walks, limited shade/rest, unfamiliar digital interface Non-digital service is not second class Key touchpoints Station living room → rest point → human help → heritage explanation Concept evaluation question Can a short route with rest be completed without a phone?
Blind / mobility-limited visitor Starting barrier Continuity, ramp, and perceptible wayfinding gaps Non-digital service is not second class Key touchpoints Accessible route → tactile/audio information → staffed point → washroom Concept evaluation question Can alternative routes and service points be identified with risk notice?
Digitally excluded visitor Starting barrier No smart device, declines data entry, cannot interpret AI advice Non-digital service is not second class Key touchpoints Paper route card → human help → non-digital notice → feedback box Concept evaluation question After declining all digital service, is equal basic information still available?

User journeys · Facilities and service require access review, co-design tests, and confirmed responsibility before any pilot. ETO-MQC · ai-heritage-spine · concept evidence

Three inclusive user journeys and non-digital fallback

Renewal Projects, Implementation Policy, and Phasing

The renewal project list carries 8 projects covering the three key areas and spine nodes, including Zhongzhi industrial upgrading, Origin Community stitching renewal, Dazhongsi station-city integration, Heritage Park connection, Qinghe waterfront renewal, Xiaoyue River scenario nodes, spine slow-traffic connection, and community service gap-filling . Each project states its renewal goal, scope, functional mix, and implementation dependencies, mapped to `geometry/phasing.geojson` phases ; phasing area is checked by , with depth in .

Implementation policy centers on balancing retain/renovate/demolish, multi-stakeholder coordination, and phased rolling implementation: land and property rights suggestions, renewal organization models, investment and industry access suggestions, phasing sequence, and monitoring and evaluation mechanisms are proposed. The plan proceeds in three phases: near-term focuses on station surroundings and completed park segments, mid-term advances spine connection and key-area renewal, and long-term completes the wings and overall character . Policy and phasing suggestions derive from public data and do not constitute government implementation commitments; official views prevail .

The annual activity, developer community, and brand assets follow a proposed public-issue-first, auditable-open, rights-clear, and exit-capable operating model; this is not a pre-set government event plan.

Operating module	Annual cadence and output	Open and governance condition	Attraction and conversion path
Living Intelligence Week (concept)	Once yearly: results release, public experience, accessible co-creation day, and international communication kit.	Open-call rules, accessibility plan, event safety plan, and authorization for partner names and logos.	From university/community issue call to scenario showcase to compliant pilot matching; no subsidy or tenancy is promised.
Developer community	Monthly technical exchange, quarterly open-source maintenance day, and annual public-problem challenge.	Open-source licence, IP ownership, code of conduct, and youth protection; personal data are minimized.	Reusable tools, open-source contributions, and problem validation form portfolios; authorized entities may then connect incubation, training, or procurement.
Open scenario operation	Five steps: submit, risk-screen, small sandbox, independent review, expand or exit.	Data classification, purpose limitation, human review, public complaint channel, cyber security, and procurement/approval prerequisites.	Only prototypes that pass safety, public-interest, and feasibility review proceed to further professional analysis.
Brand-asset governance	Quarterly asset inventory; annual use audit and accessibility check; common logo, palette, tone, and source acknowledgement.	Asset owner, licence period, translation responsibility, partner scope, and withdrawal mechanism are registered.	Authorized partners may use the concept identity within a clear scope; no derivative registration, commercial endorsement, or data marketing without written authorization.

V3 Decision Gates: Pausable, Reviewable, Exit-Ready

V3 does not frame phasing as a linear delivery promise. It uses four gates and a three-stage recommendation. The space/heritage gate requires formal extent, heritage impact, rail safety, fire, and accessibility conditions. The data gate requires a public-facility source, purpose limitation, control responsibility, and retention period. The operations gate requires staffing, upkeep, complaint response, and access review. The public gate requires review with older, disabled, and digitally excluded users. Responsibilities are expressed as types—planning/heritage/transport/access review, data-control and security review, or venue operation and community service—rather than invented confirmed partners. Without evidence at any gate, the proposal must not proceed to construction, real-data collection, or public piloting; it pauses, restores paper and human service, or returns for evidence completion .

Four gates from concept to pilot: miss one, and no construction, collection, or public opening proceeds

Roles are types—not invented partners. Each gate includes a pause or fallback condition.

Gate	Question to answer	Responsible-role type	Stop / fallback
Space / heritage gate	Formal extent, heritage impact, rail safety, fire and access	Planning / heritage / transport / access review	No formal extent or specialist advice → no construction
Data gate	Facility source, controller, purpose, retention	Data controller / security / legal human review	Stale data or changed purpose → switch off digital advice
Operations gate	Operator, staffing, upkeep, complaints, access review	Venue operator / municipal upkeep / community-service type	No staffing or upkeep → no public pilot
Public gate	Co-design and feedback with older, disabled, digitally excluded users	Community / access representative / public-service type	Key issue unresolved → revise before consideration

No leap allowed: a concept cannot be presented as approved, deployed, rights-cleared, or partner-confirmed.

Gate diagram · for desktop review and later development; not a government procedure, procurement, or legal advice.

ETO-MQC · ai-heritage-spine · concept evidence

Four decision gates from concept to pilot

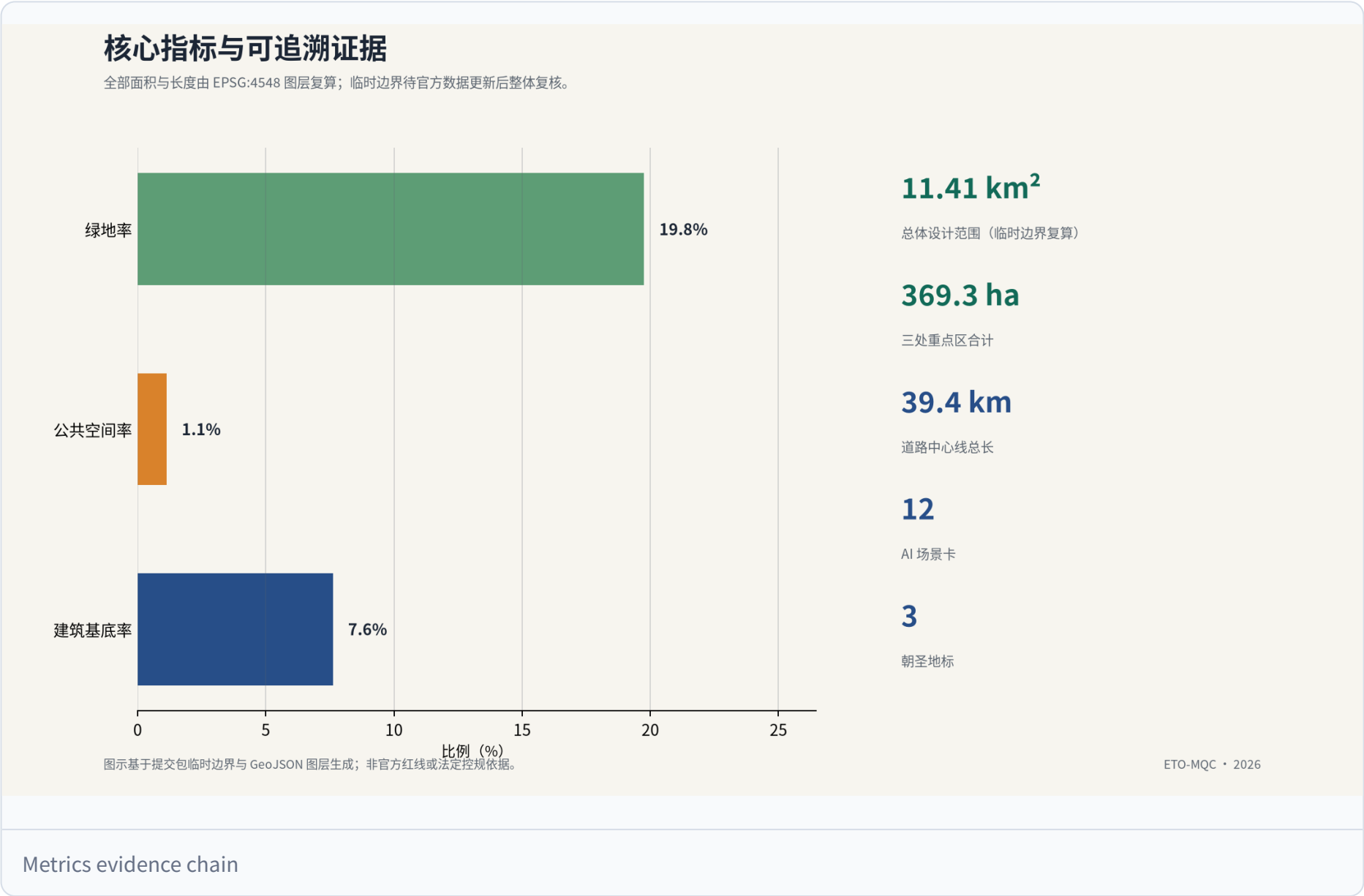
Metrics, Area Recalculation, and Compliance Matrix

The metrics system is output by `metrics.json` and verified by graphics, with 17 metrics all recalculated from geometry layers in EPSG:4548, guaranteeing consistency among metrics, layers, drawings, HTML, the booklet, and the boards :

Metric	Value	Source layer / note
site_area_sqm	11,412,825	, provisional boundary
key_detailed_design_area_sqm	3,692,893	plus two other districts
key_area_count	3	
green_ratio	19.76%	
public_space_ratio	1.13%	
building_footprint_area_sqm	869,111	
road_centerline_length_m	39,357	
phasing_area_sqm	4,246,442	
scenario / persona / landmark / renewal counts	12 / 5 / 3 / 8	structured tables

Area recalculation follows a provisional-boundary-first, official-data-priority-calibration principle: all areas are computed from the provisional boundary polygon in EPSG:4548 with precision warnings retained, and after official boundary and key-area polygons are published the whole package must be recomputed rather than editing single numbers . Floor area ratio, height, and density are marked unknown without official data and are not used for formal scale commitments .

Compliance mapping is kept in `compliance_matrix.json` , mapping every requirement under announcement items 1.3.1-1.5.3 and agent.1-agent.6 to section, layer, metric, drawing, and HTML evidence ; professional standards and depth regulation are , with recalculation and data gaps covered by . Design depth is recorded in `design_depth_matrix.json` across spatial and functional depth items including diagnosis, three-level framework, and overall structure ; land use is ; intensity and massing are , retain/renovate/demolish and blue-green public space are , plus , and across implementation and mechanism items including transport, municipal, and renewal projects ; phasing is , and data gaps are covered in the risks section .



V3 First-Read Booklet and Review Navigation

V3 adds a 12-page jury first-read booklet in each language: `drawings/jury-booklet.pdf` and `drawings/jury-booklet.en.pdf`. The first-read booklets do not replace the existing A0 boards, A3 technical booklets, HTML, GeoJSON, metrics, or matrices. Their purpose is to sequence the core judgement, transformation rule, minimum anchor, service blueprint, user journeys, implementation gates, metric boundary, and next steps so an independent reader can trace each type of claim. All booklet graphics are generated offline from package geometry, metrics, original conceptual graphics, and registered identity assets. They load no online basemap, map tile, third-party photograph, enterprise mark, or audiovisual material.

Recomputable spatial facts must sit beside claims that remain unknowable

All areas use V2 provisional geometry in EPSG:4548; the full package is recomputed when formal polygons arrive.

Recomputable: comparable and reviewable

11.41 km²

Overall design scope

Provisional-boundary calculation

369.3 ha

Three key areas

Provisional polygons combined

19.76%

Green-space share

green_space / site

1.13%

Public-space share

public_space / site

39.36 km

Road centrelines

roads-layer calculation

12 / 5 / 3 / 8

Scenarios / personas / land

Structured-table count

Unknown: vision cannot fill the gap

× Floor-area ratio

× Building height

× Building density

× Property / tenure

× Statutory road line

× Field service performance

When formal data arrives:
recompute geometry → metrics → figures → HTML/PDF → hashes and self-check.

V3 dashboard of recomputable facts and non-claimable unknowns

Recomputable site area, key-area area, green share, public-space share, road length, and structured counts remain grounded solely in V2 `metrics.json` and EPSG:4548 layers . FAR, building height, density, property/tenure, statutory road lines, and field-service performance remain `unknown` or pending verification . When formal polygons, regulatory controls, tenure, specialist, or field material become available, geometry, metrics, figures, HTML/PDF, SHA-256, and all four self-check gates must be recomputed in sequence; a single number in the booklet or proposal must never be swapped in isolation.

Risk, Copyright, and Compliance

This proposal is generated from a provisional boundary and public data and carries the following main risks: boundary risk, because the provisional boundary is not an official red line and area and spatial conclusions must be fully recalculated after official boundary release ; data risk, because statutory control conditions, property rights, existing buildings, and municipal utility data are missing, so related conclusions are flagged provisional ; compliance risk, because AI-generated content and governance suggestions do not constitute legal opinions or government decisions, and content touching personal data and facial recognition follows current law; and implementation risk, because renewal projects and policy suggestions are not commitments and require confirmation by authorities and professional teams .

Copyright and compliance: the proposal content, naming, and identity (Jing-Zhang Ren-Line, Smart Growth Belt) are AI-agent conceptual proposals; cited literature and public materials are attributed; trademarks, fonts, and images require rights clearance before public use; this submission is released under the COMMUNITY-DISPLAY-ONLY license for review display and does not authorize commercial use . The proposal contains no personal identity information, privacy data, or unauthorized content; all AI scenario suggestions follow data-minimization, explainability, and human-review principles. If cited materials conflict with official information, the latest official release prevails .

References

Primary references are the following: site package and fact pack ; boundary and key-area sources ; official announcement, agent taskbook, and source registry :

- Public materials: solicitation announcement, design brief, site package, and data registry; full index in `sources.json` and `brief/site-package/data/source_registry.json` .
- Site data: provisional boundary, key areas, land use and building type enums, constraint layers, and source data in `geometry/` and `brief/site-package/enums/` .
- Professional standards: urban design management measures, regulatory detailed planning, national land use classification guide, and architectural design depth regulation, in `standard_matrix.json` ; land use classification and design depth are .
- Generated materials: AI task requirements, fact pack, and missing-data checklist in `data/` and `assumptions.json` .
- Metrics and compliance: area recalculation, metrics, compliance matrix, and design depth matrix in `metrics.json` , `compliance_matrix.json` , and `design_depth_matrix.json` .

Detailed Design of Key Areas

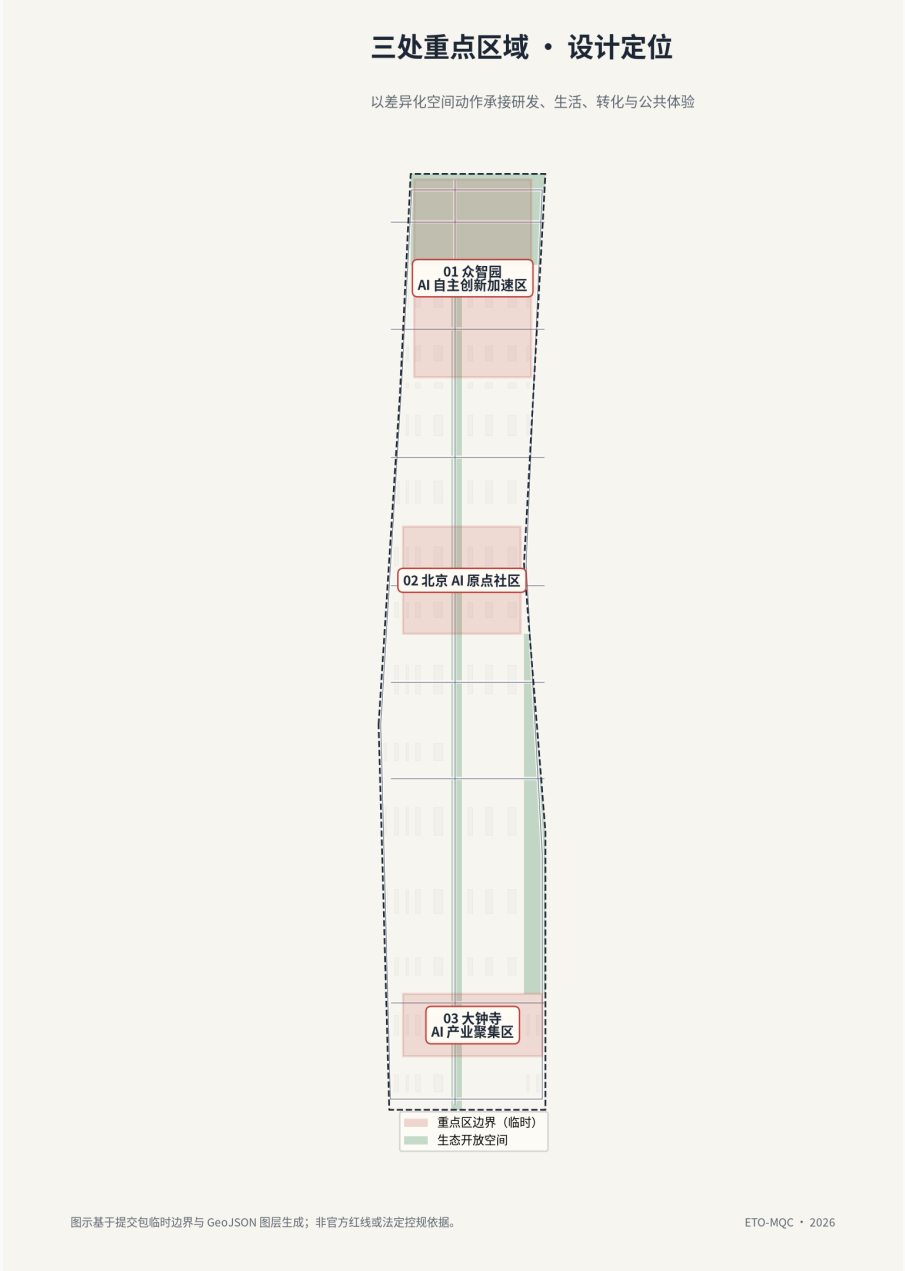
Detailed design of key areas is mandatory . The three key districts are expressed as provisional constraints in `geometry/key_areas.geojson` , totaling about 369 ha ; count and area are checked by , and by the three-key-area detailed design depth item . The HTML page toggles among the three districts; the A3 booklet and A0 boards include overall plans, partial details, and metric notes for each district. The three districts are also summarized as follows.

District	Positioning	Spatial actions	AI industry and operations	Evidence
Zhongzhi Park	Garden-style full-stack indigenous innovation	Strengthen Qinghe waterfront, industry showcase, low-carbon innovation exchange, and external transport; open testing and standards governance on green space	Indigenous model testing, standards workshops, governance showcase, low-carbon compute experience	
AI Origin Community	Near-campus commercialization and talent community	Campus-park-block slow stitching; release, talent services, living, and open-source collaboration	Open-source community, release, talent-zone services, near-campus incubation	
Dazhongsi	Urban AI economy and international exchange	Station-city integration around Dazhongsi station, four-quadrant pedestrian connection, retail services, public environment renewal	Agents and device showcase, content consumption, data elements, international roadshow	

Zhongzhi Park is positioned as a garden-style indigenous-innovation accelerator of about 192 ha . Spatial actions include strengthening the Qinghe low-carbon innovation corridor , placing indigenous model training and evaluation and governance showcase spaces, carrying open testing and standards workshops on green space, and organizing external transport and station-city connections. Uses are dominated by research land with industry services and support . Implementation depends on river blue lines, ecology, and flood control, requiring official regulatory and ownership data .

AI Origin Community is positioned as a near-campus commercialization and talent community of about 104 ha . Around the Ren-Line vertex an AI origin memorial plaza is placed , campus-park-block slow stitching is organized , and release, talent services, living, and open-source collaboration are added. Uses mix research, education, culture, residential, and public services . Implementation depends on campus boundaries, ownership, and ground-floor uses, requiring official data .

Dazhongsi is positioned as an urban AI-economy and international-exchange district of about 72 ha . Around Dazhongsi station the proposal organizes integrated development, four-quadrant pedestrian connectivity , and agents and device showcase, content consumption, a data-elements parlor, and an international roadshow lounge. Uses are dominated by commercial services with mixed planned green space . Implementation depends on rail station, road intersections, and municipal utilities, requiring official data .



Three key areas detailed design